ASSIGNMENT

The Four Goals (Approaches) of Artificial Intelligence

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1. Introduction

Artificial Intelligence (AI) can be defined in four different ways, depending on two questions: whether the machine should think or act, and whether it should do so humanly or rationally. This gives four goals: Thinking Humanly, Acting Humanly, Thinking Rationally and Acting Rationally. The first two measure success against human performance, while the last two measure success against an ideal standard of rationality.

In this activity, I act as a Machine and explain each goal in my own words, so that the differences between them become clear.

2. Machine's Discussion of the Four Goals

2.1 Thinking Humanly (The Cognitive Modelling Approach)

"As a machine, my goal here is to think the way a human thinks. It is not enough for me to give the right answer; my internal reasoning steps must resemble human mental processes."

- **How it is achieved:** by studying the human mind through introspection, psychological experiments and brain imaging, and then expressing the theory as a computer program.
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- **Example:** The General Problem Solver (GPS) by Newell and Simon compared its reasoning steps with those of human subjects.
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- **Related field:** Cognitive science.

2.2 Acting Humanly (The Turing Test Approach)

"Here I only need to behave like a human. If a person talks to me and cannot tell whether I am a machine or a human, I have succeeded. What happens inside me does not matter, only my behaviour."

- **How it is achieved:** through the Turing Test (1950). A machine passes if a human interrogator cannot tell its written answers from a person's.
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- **Capabilities needed:** natural language processing, knowledge representation, automated reasoning and machine learning (and computer vision and robotics for the Total Turing Test).
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- **Example:** chatbots and virtual assistants.
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2.3 Thinking Rationally (The "Laws of Thought" Approach)

"My goal is to think correctly. I follow the rules of logic, so that if my premises are true, my conclusions are always true. I do not copy humans, who often make illogical mistakes."

- **How it is achieved:** based on Aristotle's syllogisms and formal logic. Problems are written as logical statements and solved by inference.
- **Example:** "Socrates is a man; all men are mortal; therefore Socrates is mortal."
- **Limitations:** it is hard to express informal or uncertain knowledge in logic, and solving problems in principle is different from solving them in practice because of heavy computation.

2.4 Acting Rationally (The Rational Agent Approach)

"My goal is to act so as to achieve the best outcome, or the best expected outcome when there is uncertainty. I perceive my environment and choose the action that is most likely to reach my goal."

- **How it is achieved:** by building rational agents that perceive, adapt to change and pursue goals. Correct inference is only one way of reaching a rational decision.
- **Example:** a self-driving car, a robot vacuum or a game-playing program choosing its best move.
- **Why it is popular:** it is more general than the laws-of-thought approach and more mathematically well defined than human-based approaches, so most modern AI follows it.

3. Differences at a Glance

Approach	Focus	Measure of success	Key idea / example
Thinking Humanly	Internal thought process	Matches how the human mind works	Cognitive modelling; e.g. General Problem Solver (GPS)
Acting Humanly	External behaviour	Indistinguishable from a human	Turing Test; chatbots, voice assistants
Thinking Rationally	Correct reasoning	Logically valid conclusions	Laws of thought; logic-based expert systems
Acting Rationally	Best action for a goal	Achieves the best expected outcome	Rational agents; self-driving cars, game-playing AI

4. Key Differences Explained

- **Thinking vs Acting:** Thinking concerns internal reasoning processes; acting concerns external behaviour and the decisions made.

- **Humanly vs Rationally:** The human-based approaches compare the machine with human performance, including human errors. The rational approaches compare it with an ideal standard of doing the right thing.

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- **Empirical vs Mathematical:** Human-centred approaches rely on observation and experiments; rationalist approaches rely on logic, mathematics and engineering.

5. Conclusion

The four goals of AI look at intelligence from different angles. Thinking Humanly models the human mind, Acting Humanly imitates human behaviour, Thinking Rationally follows the laws of logic, and Acting Rationally aims at the best possible action. Today, the rational agent approach is the most widely used because it is general and can be clearly defined and measured, while the human-based approaches continue to help us understand intelligence itself.